

## Assignment-1.2

### Submitted by : Yogesh Indoria

- Create a utility(FileManager.sh) that will be able to
- Create a Directory
- Delete a Directory
- List Content of a Directory
- Only listfiles in a Directory
- Only listDirs in a Directory
- listAll(files and directory)

```
yogesh@DEVICE-251579:~$ touch FileManager.sh
yogesh@DEVICE-251579:~$ nano FileManager.sh
yogesh@DEVICE-251579:~$ chmod +x FileManager.sh
yogesh@DEVICE-251579:~$ ls_l
ls_l: command not found
yogesh@DEVICE-251579:~$ ls -l
total 24
drwxr-xr-x 4 yogesh yogesh 4096 Jul 29 08:11 Dir1
drwxr-xr-x 2 yogesh yogesh 4096 Jul 29 08:07 Dir2
drwxr-xr-x 2 yogesh yogesh 4096 Jul 29 08:07 Dir3
drwxr-xr-x 2 yogesh yogesh 4096 Jul 29 08:07 Dir4
-rwxr-xr-x 1 yogesh yogesh 3312 Jul 29 09:49 FileManager.sh
-rwxr-xr-x 1 yogesh yogesh 1194 Jul 29 08:03 yog1.sh
```

```
yogesh@DEVICE-251579:~$ ./FileManager.sh addDir /tmp dir1
Directory 'dir1' created successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh addDir /tmp dir2
Directory 'dir2' created successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh addDir /tmp dir3
Directory 'dir3' created successfully.
```

```
yogesh@DEVICE-251579:~$ ./FileManager.sh listFiles /tmp
/tmp/file1.txt
/tmp/file2.txt
yogesh@DEVICE-251579:~$ ./FileManager.sh listDirs /tmp
/tmp/systemd-private-35c689607818433db8c021e1a7e1f0e4-systemd-logind.service-BbwsSY
/tmp/systemd-private-35c689607818433db8c021e1a7e1f0e4-systemd-timesyncd.service-ZbbdVm
/tmp/dir5
/tmp/systemd-private-35c689607818433db8c021e1a7e1f0e4-polkit.service-rMk1k8
/tmp/dir1
/tmp/systemd-private-35c689607818433db8c021e1a7e1f0e4-systemd-resolved.service-5GApQo
/tmp/.X11-unix
/tmp/snap-private-tmp
/tmp/dir2
/tmp/dir3
yogesh@DEVICE-251579:~$ ./FileManager.sh listAll /tmp
total 44
drwxrwxrwt 12 root root 4096 Jul 29 10:04 .
drwxr-xr-x 22 root root 4096 Jul 29 06:52 ..
drwxrwxrwx 2 root root 60 Jul 29 06:52 .X11-unix
drwxr-xr-x 2 yogesh yogesh 4096 Jul 29 09:55 dir1
drwxr-xr-x 2 yogesh yogesh 4096 Jul 29 09:55 dir2
drwxr-xr-x 2 yogesh yogesh 4096 Jul 29 09:55 dir3
drwxr-xr-x 2 yogesh yogesh 4096 Jul 29 09:54 dir5
-rw-r--r-- 1 yogesh yogesh 0 Jul 29 10:04 file1.txt
-rw-r--r-- 1 yogesh yogesh 0 Jul 29 10:04 file2.txt
drwx----- 2 root root 4096 Jul 29 06:52 snap-private-tmp
drwx----- 3 root root 4096 Jul 29 07:29 systemd-private-35c689607818433db8c021e1a7e1f0e4-polkit.service-rMk1k8
drwx----- 3 root root 4096 Jul 29 06:52 systemd-private-35c689607818433db8c021e1a7e1f0e4-systemd-logind.service-BbwsSY
drwx----- 3 root root 4096 Jul 29 06:52 systemd-private-35c689607818433db8c021e1a7e1f0e4-systemd-resolved.service-5GApQo
drwx----- 3 root root 4096 Jul 29 06:52 systemd-private-35c689607818433db8c021e1a7e1f0e4-systemd-timesyncd.service-ZbbdVm
yogesh@DEVICE-251579:~$ ./FileManager.sh listContent /tmp
dir1 file1.txt
dir2 file2.txt
dir3 snap-private-tmp
dir5 systemd-private-35c689607818433db8c021e1a7e1f0e4-polkit.service-rMk1k8
systemd-private-35c689607818433db8c021e1a7e1f0e4-systemd-logind.service-BbwsSY
systemd-private-35c689607818433db8c021e1a7e1f0e4-systemd-resolved.service-5GApQo
systemd-private-35c689607818433db8c021e1a7e1f0e4-systemd-timesyncd.service-ZbbdVm
yogesh@DEVICE-251579:~$
```

Update FileManager to process Files as well

- Create a file
- Add content to file
- Add content at the beginning of the file
- Show top n lines of a file
- Show last n lines of a file
- Show contents of a specific line number
- Show content of a specific line number range
- Move/Copy file from one location to another
- Delete file

```

yogesh@DEVICE-251579:~$ ./FileManager.sh addFile /tmp/dir1 file1.txt
File 'file1.txt' created successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh addFile /tmp/dir1 file1.txt "Initial Content"
File 'file1.txt' created successfully with initial content.
yogesh@DEVICE-251579:~$ ./FileManager.sh addContentToFile /tmp/dir1 file1.txt "Additional Content"
Content appended successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh addContentToFileBeginning /tmp/dir1 file1.txt "Additional Content"
Content added at beginning successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh showFileBeginningContent /tmp/dir1 file1.txt 5
Additional Content
Initial Content
Additional Content
yogesh@DEVICE-251579:~$ ./FileManager.sh showFileEndContent /tmp/dir1 file1.txt 5
Additional Content
Initial Content
Additional Content
yogesh@DEVICE-251579:~$ ./FileManager.sh showFileEndContent /tmp/dir1 file1.txt 1
Additional Content
yogesh@DEVICE-251579:~$ ./FileManager.sh showFileBeginningContent /tmp/dir1 file1.txt 2
Additional Content
Initial Content
yogesh@DEVICE-251579:~$ ./FileManager.sh addContentToFile /tmp/dir1 file1.txt "Line 1"
Content appended successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh addContentToFile /tmp/dir1 file1.txt "Line 2"
Content appended successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh addContentToFile /tmp/dir1 file1.txt "Line 3"
Content appended successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh addContentToFile /tmp/dir1 file1.txt "Line 4"
Content appended successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh addContentToFile /tmp/dir1 file1.txt "Line 5"
Content appended successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh addContentToFile /tmp/dir1 file1.txt "Line 6"
Content appended successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh addContentToFile /tmp/dir1 file1.txt "Line 7"
Content appended successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh addContentToFile /tmp/dir1 file1.txt "Line 8"
Content appended successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh addContentToFile /tmp/dir1 file1.txt "Line 9"
Content appended successfully.

yogesh@DEVICE-251579:~$ ./FileManager.sh showFileEndContent /tmp/dir1 file1.txt 5
Line 6
Line 7
Line 8
Line 9
Line 10
yogesh@DEVICE-251579:~$ ./FileManager.sh showFileContentAtLine /tmp/dir1 file1.txt 10
Line 7
yogesh@DEVICE-251579:~$ ./FileManager.sh showFileContentForLineRange /tmp/dir1 file1.txt 5 10
Line 2
Line 3
Line 4
Line 5
Line 6
Line 7
yogesh@DEVICE-251579:~$ ./FileManager.sh moveFile /tmp/dir1/file1.txt /tmp/dir1/file2.txt
File moved successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh moveFile /tmp/dir1/file2.txt /tmp/dir2/
File moved successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh copyFile /tmp/dir2/file2.txt /tmp/dir1/
File copied successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh copyFile /tmp/dir1/file2.txt /tmp/dir1/file3.txt
File copied successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh clearFileContent /tmp/dir1 file3.txt
File content cleared successfully.
yogesh@DEVICE-251579:~$ ./FileManager.sh deleteFile /tmp/dir1 file2.txt
File deleted successfully.
yogesh@DEVICE-251579:~$

```

## [FileManager.sh](#) File

```
#!/bin/bash
```

```
# FileManager.sh Executable File
```

```
# Command is provided
```

```
if [ $# -lt 1 ]; then
```

```
    echo "Error: No command provided."
```

```
    echo ""
```

```
    echo "Usage:"
```

```
    echo "  ./FileManager.sh addDir <path> <dirname>"
```

```
    echo "  ./FileManager.sh deleteDir <path> <dirname>"
```

```
    echo "  ./FileManager.sh listContent <path>"
```

```
    echo "  ./FileManager.sh listFiles <path>"
```

```
    echo "  ./FileManager.sh listDirs <path>"
```

```
    echo "  ./FileManager.sh listAll <path>"
```

```
    exit 1
```

```
fi

case "$1" in

# Create Directory

addDir)

    if [ $# -ne 3 ]; then
        echo "Usage: ./FileManager.sh addDir <path> <directory_name>"
        exit 1
    fi

    if [ ! -d "$2" ]; then
        echo "Error: Directory '$2' does not exist."
        exit 1
    fi

    mkdir -p "$2/$3"

    echo "Directory '$3' created successfully."
    ;;

# Delete Directory
# ./FileManager.sh deleteDir /tmp dir1

deleteDir)

    if [ $# -ne 3 ]; then
        echo "Usage: ./FileManager.sh deleteDir <path> <directory_name>"
        exit 1
    fi

    if [ ! -d "$2/$3" ]; then
        echo "Error: Directory '$2/$3' does not exist."
        exit 1
    fi

    rm -r "$2/$3"

    echo "Directory '$3' deleted successfully."
    ;;
```

```
# List Content of Directory
# ./FileManager.sh listContent /tmp
```

```
listContent)
```

```
if [ $# -ne 2 ]; then
    echo "Usage: ./FileManager.sh listContent <path>"
    exit 1
fi
```

```
if [ ! -d "$2" ]; then
    echo "Error: Directory '$2' does not exist."
    exit 1
fi
```

```
ls "$2"
;;
```

```
# List Only Files
# ./FileManager.sh listFiles /tmp
```

```
listFiles)
```

```
if [ $# -ne 2 ]; then
    echo "Usage: ./FileManager.sh listFiles <path>"
    exit 1
fi
```

```
if [ ! -d "$2" ]; then
    echo "Error: Directory '$2' does not exist."
    exit 1
fi
```

```
find "$2" -maxdepth 1 -type f
;;
```

```
# List Only Directories
# ./FileManager.sh listDirs /tmp
```

```
listDirs)
```

```
if [ $# -ne 2 ]; then
```

```

        echo "Usage: ./FileManager.sh listDirs <path>"
        exit 1
    fi

    if [ ! -d "$2" ]; then
        echo "Error: Directory '$2' does not exist."
        exit 1
    fi

    find "$2" -mindepth 1 -maxdepth 1 -type d
    ;;

# List All Files & Directories
# ./FileManager.sh listAll /tmp

listAll)

    if [ $# -ne 2 ]; then
        echo "Usage: ./FileManager.sh listAll <path>"
        exit 1
    fi

    if [ ! -d "$2" ]; then
        echo "Error: Directory '$2' does not exist."
        exit 1
    fi

    ls -la "$2"
    ;;

# Add or Create File
# ./FileManager.sh addFile /tmp/dir1 file1.txt

addFile)

    if [ $# -lt 3 ] || [ $# -gt 4 ]; then
        echo "Usage: ./FileManager.sh addFile <path> <filename> [content]"
        exit 1
    fi

    if [ ! -d "$2" ]; then
        echo "Error: Directory '$2' does not exist."
        exit 1
    fi

```

```
if [ $# -eq 3 ]; then
    touch "$2/$3"
    echo "File '$3' created successfully."
else
    echo "$4" > "$2/$3"
    echo "File '$3' created successfully with initial content."
fi

;;
```

```
# Add content to file
# ./FileManager.sh addContentToFile /tmp/dir1 file1.txt Additional Content
```

```
addContentToFile)
```

```
if [ $# -ne 4 ]; then
    echo "Usage: ./FileManager.sh addContentToFile <path> <filename> <content>"
    exit 1
fi
```

```
if [ ! -f "$2/$3" ]; then
    echo "Error: File does not exist."
    exit 1
fi
```

```
echo "$4" >> "$2/$3"
```

```
echo "Content appended successfully."
```

```
;;
```

```
# Add Content to Beginning of File
# ./FileManager.sh addContentToFileBeginning /tmp/dir1 file1.txt "Additional Content"
```

```
addContentToFileBeginning)
```

```
if [ $# -ne 4 ]; then
    echo "Usage: ./FileManager.sh addContentToFileBeginning <path> <filename> <content>"
    exit 1
fi
```

```
if [ ! -f "$2/$3" ]; then
    echo "Error: File does not exist."
```

```

        exit 1
    fi

    {
        echo "$4"
        cat "$2/$3"
    } > "$2/$3.tmp"

    mv "$2/$3.tmp" "$2/$3"

    echo "Content added at beginning successfully."

    ;;

# Show top n lines of a file
# ./FileManager.sh showFileBeginningContent /tmp/dir1 file1.txt 5

showFileBeginningContent)

    if [ $# -ne 4 ]; then
        echo "Usage: ./FileManager.sh showFileBeginningContent <path> <filename> <lines>"
        exit 1
    fi

    head -n "$4" "$2/$3"

    ;;

# Show last n lines of a file
# ./FileManager.sh showFileEndContent /tmp/dir1 file1.txt 5

showFileEndContent)

    if [ $# -ne 4 ]; then
        echo "Usage: ./FileManager.sh showFileEndContent <path> <filename> <lines>"
        exit 1
    fi

    tail -n "$4" "$2/$3"

    ;;

# Show contents of a specific line number
# ./FileManager.sh showFileContentAtLine /tmp/dir1 file1.txt 10

```

showFileContentAtLine)

```
if [ $# -ne 4 ]; then
    echo "Usage: ./FileManager.sh showFileContentAtLine <path> <filename> <line_number>"
    exit 1
fi
```

```
sed -n "${4}p" "$2/$3"
```

```
::
```

#Show conteint of a specific line number range

```
# ./FileManager.sh showFileContentForLineRange /tmp/dir1 file1.txt 5 10
```

showFileContentForLineRange)

```
if [ $# -ne 5 ]; then
    echo "Usage: ./FileManager.sh showFileContentForLineRange <path> <filename> <start_line>
<end_line>"
    exit 1
fi
```

```
sed -n "${4},${5}p" "$2/$3"
```

```
::
```

# Move File

```
# ./FileManager.sh moveFile /tmp/dir1/file1.txt /tmp/dir1/file2.txt
```

moveFile)

```
if [ $# -ne 3 ]; then
    echo "Usage: ./FileManager.sh moveFile <source> <destination>"
    exit 1
fi
```

```
mv "$2" "$3"
```

```
echo "File moved successfully."
```

```
::
```

# Copy File

```
# ./FileManager.sh copyFile /tmp/dir2/file2.txt /tmp/dir1/
```

```
copyFile)
```

```
if [ $# -ne 3 ]; then
    echo "Usage: ./FileManager.sh copyFile <source> <destination>"
    exit 1
fi
```

```
cp "$2" "$3"
```

```
echo "File copied successfully."
```

```
::
```

```
# Delete file
```

```
# ./FileManager.sh deleteFile /tmp/dir1 file2.txt
```

```
deleteFile)
```

```
if [ $# -ne 3 ]; then
    echo "Usage: ./FileManager.sh deleteFile <path> <filename>"
    exit 1
fi
```

```
rm "$2/$3"
```

```
echo "File deleted successfully."
```

```
::
```

```
# Clear File Content
```

```
# ./FileManager.sh clearFileContent /tmp/dir1 file3.txt
```

```
clearFileContent)
```

```
if [ $# -ne 3 ]; then
    echo "Usage: ./FileManager.sh clearFileContent <path> <filename>"
    exit 1
fi
```

```
> "$2/$3"
```

```
echo "File content cleared successfully."
```

```
;;
```

```
# Invalid Command
```

```
*)
```

```
    echo "Invalid Command: $1"
```

```
    echo ""
```

```
    exit 1
```

```
;;
```

```
esac
```